

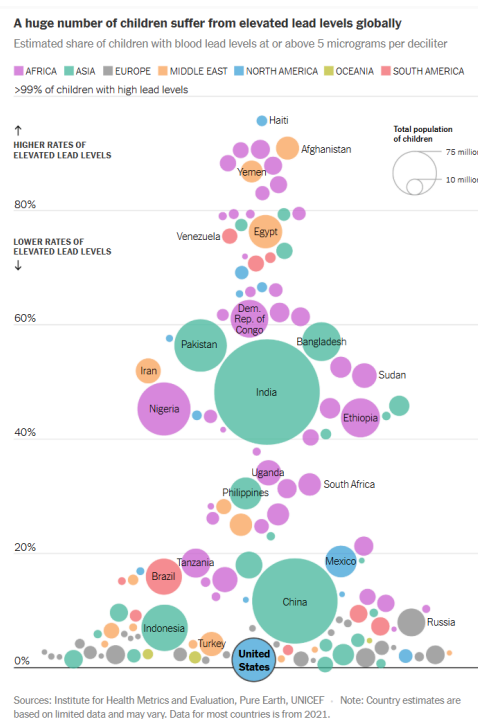
Homework 01

Insert your name

Answer each of the following questions. Submit both the rendered PDF file and the source .qmd file. Explanations/analyses should be polished writing (complete sentences, good grammar, clear reasoning, etc)

1. Analyze the following figure. (Taken from “[Will This New U.S. Project Make the World’s Children Smarter?](#)” (N. Kristoff, Oct 23, 2024, NY Times)

1. Identify the cases, variables, and visual cues being used.
2. Describe and critique the graphic in terms of ideas from Chapter 2. (Think about the scale being used, the use of color, labels and other text, etc.)
3. Describe, in words, what *information* you think the data graphic conveys. Do not just summarize the *data*—interpret the data in context and tell us what it means.



2. (Chapter 3, #7) The `RailTrail` data set from the `mosaicData` package describes the usage of a rail trail in Western Massachusetts. Use these data to create a plot with the following characteristics.

1. Create a scatterplot of the number of crossings per day `volume` against the high temperature that day.
 2. Separate your plot into facets by `weekday` (an indicator of weekend/holiday vs. weekday).
 3. Add regression lines to the two facets.
 4. Add informative title and axis labels.
 5. Write a few sentences about what we can take away from the plot you created.
3. View the interactive graphic “US Measles Map, 2025” from the article [“Measles tracker: Follow cases, outbreaks and vaccination rates across the U.S.”](#) (Joe Murphy and Melinda Yao, April 1, 2025, NBC News). Make sure to click through all of the tabs for the map.
 1. Identify the cases, variables, and visual cues being used.
 2. You can find the underlying data for these maps [HERE](#). (I also posted [my script for scraping and cleaning the data](#), in case you are interested.) Use the data to create one clear and informative graphic about measles.
 3. Write a paragraph comparing and contrasting your graphic with the original maps. What can you learn from each? Advantages and disadvantages for the different plots?
 4. (Chapter 3, #3) This question uses the `Marriage` dataset from the `mosaicData` package.
 1. Create an informative and meaningful data graphic.
 2. Describe your choices for the graphic and why you think they are effective.
 3. Create a data graphic with at least *five* variables (either quantitative or categorical). For the purposes of this exercise, do not worry about making your visualization meaningful—just try to encode five variables into one plot.